

A disproof of Conjecture 00000000172

Under the stated definition, $\sigma = \lim \sigma_n^{1/n} = 1$

1 The conjecture and its reading

Conjecture 00000000172 reads: *Definition: Somos's constant $\sigma = \lim \sigma_n^{1/n}$, where $\sigma_n = \sigma_{n-1} + \sigma_{\lfloor n/2 \rfloor}$, $\sigma_1 = 1$. Conjecture: The density of 1's in the binary expansion of σ diverges, with counting function growing as $\Theta(\log \log n)$; this yields an explicit proof path for the irrationality of σ .* The Chinese text gives the same definition ($\sigma = \lim \sigma_n^{1/n}$, $\sigma_n = \sigma_{n-1} + \sigma_{\lfloor n/2 \rfloor}$, $\sigma_1 = 1$) and the same three claims.

Which σ . We use the definition written in the conjecture (identical in both languages). The recurrence is OEIS A033485, “ $a(n) = a(n-1) + a(\lfloor n/2 \rfloor)$, $a(1) = 1$ ” (retrieved), with terms 1, 2, 3, 5, 7, 10, 13, 18, ...; it is defined for $n \geq 1$, the recurrence applying for $n \geq 2$. We note openly that the name “Somos' constant” usually denotes a different number, Somos' quadratic recurrence constant $\sqrt{1\sqrt{2\sqrt{3}\cdots}} \approx 1.661687949633594$ (Wikipedia, retrieved); nothing below concerns that number.

The claims refuted. Writing $C(n)$ for the number of 1's among the first n binary digits of σ : (i) $C(n) = \Theta(\log \log n)$; (ii) the density $C(n)/n$ diverges (we refute both meanings: “has no limit” and “tends to $+\infty$ ”); (iii) σ is irrational (the claim that the above “yields a proof” of irrationality). We show $\sigma = 1$; then (iii) fails, and (i), (ii) fail for every binary expansion of 1.

Conventions. $\log$ is the natural logarithm; $\sigma_n^{1/n}$ is the real power of the positive number σ_n . A binary expansion of x is a pair (m, d) with $m \in \mathbb{Z}$, digits $d_k \in \{0, 1\}$ ($k \geq 0$) and $x = m + \sum_{k \geq 0} d_k 2^{-(k+1)}$. We set $C_d(n) = \#\{k < n : d_k = 1\}$ (fractional digits). To cover any convention for counting the digits of the integer part, every statement is proved for $c + C_d(n)$ with an arbitrary real offset c . Θ is Mathlib's `IsTheta` (both $f = O(g)$ and $g = O(f)$ along `atTop`).

2 Proof that $\sigma = 1$

Let $a : \mathbb{N} \rightarrow \mathbb{R}$ be *any* sequence with $a_1 = 1$ and $a_n = a_{n-1} + a_{\lfloor n/2 \rfloor}$ for $n \geq 2$ (`IsSomosSeq`; a_0 is unconstrained and never used). Such a sequence exists (`A`, `isSomosSeq_A`).

Lemma 1. (*one_le*) $a_n \geq 1$ for $n \geq 1$. (*mono*) $a_m \leq a_n$ for $1 \leq m \leq n$. (*le_mul_half*) $a_n \leq n a_{\lfloor n/2 \rfloor}$ for $n \geq 2$. (*le_pow*) $a_n \leq n^{\lfloor \log_2 n \rfloor + 1}$ for $n \geq 1$.

Proof. The first two are inductions using $a_{\lfloor n/2 \rfloor} \geq 1$. For the third, $a_2 = 2 = 2a_1$, and if $a_n \leq n a_{\lfloor n/2 \rfloor}$ then $a_{n+1} = a_n + a_{\lfloor (n+1)/2 \rfloor} \leq n a_{\lfloor (n+1)/2 \rfloor} + a_{\lfloor (n+1)/2 \rfloor}$ by monotonicity. For the fourth, by strong induction, using $\lfloor \log_2 \lfloor n/2 \rfloor \rfloor + 1 = \lfloor \log_2 n \rfloor$ for $n \geq 2$ (Mathlib `Nat.log_div_base`): $a_n \leq n \cdot \lfloor n/2 \rfloor^{\lfloor \log_2 n \rfloor} \leq n^{\lfloor \log_2 n \rfloor + 1}$. $\square$

Theorem 2 (`tendsto_root, sigma_eq_one`). $a_n^{1/n} \rightarrow 1$. Hence the conjecture's σ exists and equals 1.

Proof. Since $2^{\lfloor \log_2 n \rfloor} \leq n$, $\lfloor \log_2 n \rfloor \leq \log n / \log 2$, so by Lemma 1 (`log_le`) $0 \leq \log a_n \leq (\log n / \log 2 + 1) \log n$. Thus $0 \leq \frac{\log a_n}{n} \leq \frac{(\log n)^2}{(\log 2)n} + \frac{\log n}{n} \rightarrow 0$ (`Mathlib Real.tendsto_pow_log_div_mul_add_atTop`), and $a_n^{1/n} = \exp(\log a_n / n) \rightarrow e^0 = 1$. $\square$

3 Failure of the claims

Lemma 3 (`expansion_of_one`). If (m, d) is a binary expansion of 1, then either $d_k = 0$ for all k , or $d_k = 1$ for all k .

Proof. Let $S = \sum_k d_k 2^{-(k+1)}$; the terms are non-negative and bounded by $2^{-(k+1)}$, so $0 \leq S \leq 1$ and $m = 1 - S \in [0, 1] \cap \mathbb{Z}$. If $m = 1$ then $S = 0$, and a digit $d_k = 1$ would give $S \geq 2^{-(k+1)} > 0$. If $m = 0$ then $S = 1$, so $\sum_k (2^{-(k+1)} - d_k 2^{-(k+1)}) = 1 - 1 = 0$ with non-negative terms, and a digit $d_k = 0$ would make this sum $\geq 2^{-(k+1)} > 0$. $\square$

So $C_d(n) = 0$ for all n , or $C_d(n) = n$ for all n .

Lemma 4 (`refute`). Let $c \in \mathbb{R}$ and $F(n) = c$ for all n , or $F(n) = c + n$ for all n . Then F is not $\Theta(\log \log n)$, and $F(n)/n$ converges.

Proof. If $F \equiv c$ and $\log \log n = O(c)$, then $\log \log n$ would be eventually bounded, but $\log \log n \rightarrow \infty$. If $F(n) = c + n$ and $c + n = O(\log \log n)$, then, as $\log \log n = o(\log n)$ and $\log n = o(n)$ (`Mathlib Real.isLittleO_log_id_atTop`), $c + n = o(n)$, so eventually $|c + n| \leq n/2$, false for $n > 2|c|$. Finally $c/n \rightarrow 0$ and $(c + n)/n = 1 + c/n \rightarrow 1$. $\square$

Theorem 5 (`C172.conjecture_172_false`). Let a satisfy the definition and let $a_n^{1/n} \rightarrow \sigma$. Then $\sigma = 1$; σ is not irrational; and for every binary expansion (m, d) of σ and every $c \in \mathbb{R}$: $c + C_d(n)$ is not $\Theta(\log \log n)$; $(c + C_d(n))/n$ converges to a finite limit; and $(c + C_d(n))/n \not\rightarrow +\infty$.

Proof. Uniqueness of limits and the theorem above give $\sigma = 1$; 1 is rational (`Mathlib not_irrational_one`); the two lemmas give the rest. $\square$

Remark 6 (Scope). All three claims (i)–(iii) are refuted for every binary expansion of σ . A reading of “the density diverges” as “the number of 1’s tends to infinity” (rather than the density) is not refuted on its own for the expansion $0.111\dots$; it is refuted together with clause (i), which fails for every expansion.

4 Lean formalization

File `lean/Conjecture172/Basic.lean` (256 lines, only `import Mathlib`), namespace `C172`. Main theorem (ASCII rendering of the Unicode source):

```
theorem conjecture_172_false (a : Nat -> Real) (ha : IsSomoSeq a)
  (s : Real)
  (hs : Tendsto (fun n : Nat => a n ^ ((1 : Real) / n)) atTop (nhds s)) :
  s = 1 /\ not (Irrational s) /\
  forall (m : Int) (d : Nat -> Nat), IsBinaryExpansion s m d ->
  forall c : Real,
```

```

not ((fun n : Nat => c + (onesCount d n : Real)) =Theta[atTop]
    fun n : Nat => log (log (n : Real))) /\
(exists L, Tendsto (fun n : Nat => (c + (onesCount d n : Real)) / n)
    atTop (nhds L)) /\
not (Tendsto (fun n : Nat => (c + (onesCount d n : Real)) / n)
    atTop atTop)

```

Other theorems: `isSomosSeq_A`, `one_le`, `mono`, `le_mul_half`, `le_pow`, `log_le`, `tendsto_root`, `expansion_of_one`, `tendsto_loglog`, `loglog_isLittle0`, `refute`, `sigma_eq_one`.

Axiom audit. lake build succeeds, and `#print axioms` for `C172.conjecture_172_false`, `C172.sigma_eq_one` and `C172.isSomosSeq_A` reports only `[propext, Classical.choice, Quot.sound]`. No sorry, no `native_decide`, no added axiom.

Sources

- OEIS A033485, <https://oeis.org/A033485>: “A033485: $a(n) = a(n-1) + a(\text{floor}(n/2))$, $a(1) = 1$.”
- Wikipedia, *Somos’ quadratic recurrence constant*, https://en.wikipedia.org/wiki/Somoss%27_quadratic_recurrence_constant: “which gives a numerical value of approximately” $\sigma = 1.661687949633594121295\dots$